TryHackMe Writeup

Linux Logging for SOC

DG Interns Hub - Week 3 | **Name: Harini E | Intern ID: DG/15/OCT/CYBER/040**

Platform	TryHackMe (free room)
Room	Linux Logging for SOC
Level / path	Easy, SOC Level 1 > Linux Security Monitoring
Room time	45 min (as shown on the room page)
Status	Completed, the room progress bar shows 100%
Room link	https://tryhackme.com/room/linuxloggingforsoc

Overview

This room explains the main log sources on a Linux server and how a SOC analyst uses them while triaging an alert. It focuses on Linux servers without a desktop, such as Debian, Ubuntu, CentOS and RHEL. Most of the logs are plain text files stored in /var/log, so the work is done on the command line, as the root user, inside a prepared virtual machine. I took it because it connects directly to this week's topic: auditd and the Linux log files are the same sources I worked with in the Linux tasks.

Topics I completed

Topic	Focus
Working with text logs	Linux logs are mostly plain text files in /var/log, and they can be searched with commands like grep. One example search is `grep -R -E "auth|login|session" /var/log`.
Authentication logs	auth.log records logins, sudo use and session activity, so it is the first place to look when checking for initial access.
System and application logs	syslog and application specific logs show what services and programs were doing on the machine.
Runtime monitoring	auditd records activity while it happens, such as commands being executed, which normal text logs do not show in detail.
Bash history	History files can be edited or deleted, so they are not reliable evidence on their own.
Logging caveats	Log format, detail and location can change between distributions, so a log may look different or may not exist at all.

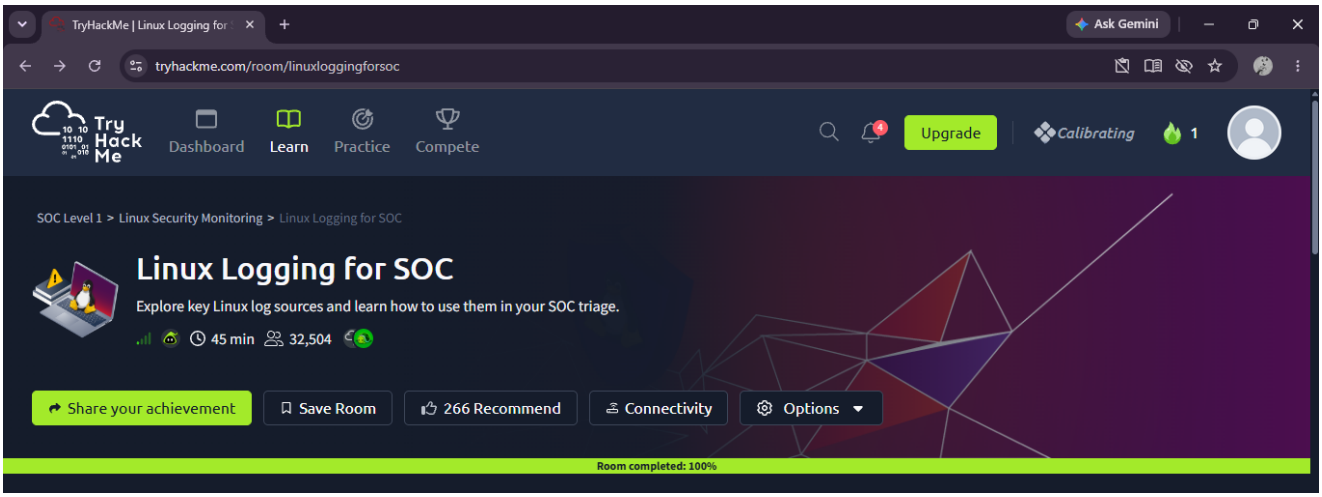
Log files to remember

Log file	What it helps me find
/var/log/auth.log	Logins, failed passwords, sudo use and session changes.
/var/log/syslog	General system and service messages.
/var/log/audit/audit.log	Detailed auditd records, including executed commands.

What I took from it

The main thing I learned is how to start a Linux investigation. I search /var/log for words like auth, login and session, read the authentication log to see who logged in and who used sudo, and then use auditd when I need to know exactly which commands were run. I also learned not to depend on bash history alone, and to expect that logs differ from one system to another. This matches this week's work with auditd, where I searched the audit log for a command I had run myself.

Evidence



1. Room page showing Linux Logging for SOC completed, room progress 100%